

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester IV)

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AIM(A): Create a simple calculator application using servlet.

Code html:-

```
<html>
  <head>
    <title>Calculator</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initialscale=1.0">
  </head>
  <body>
    <form action="Calculator" method="get" name="frm">
      Enter Number 1 :
      <input name="num1" type="text"/></br>
      Enter Number 2 :
      <input name="num2" type="text"/></br>
      Operator
      <select name="Operator">
        <option value="Addition">+</option>
        <option value="Subtraction">-</option>
        <option value="Multiplication">*</option>
        <option value="Division">/</option>
      </select>
      <input type="reset" value="reset"/>
      <input type="Submit" value="submit"/>
    </form>
  </body>
</html>
```

Code Servlet:-

```
import static java.io.FileDescriptor.out;
import java.io.IOException;
import java.io.PrintWriter;
import static java.lang.System.out;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
```

```
@WebServlet(urlPatterns = {"/Calculator"})
public class calculator extends HttpServlet {
    protected void processRequest(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
        PrintWriter out=response.getWriter();
        String num1= request.getParameter("num1");
```

```

String num2= request.getParameter("num2");
String op = request.getParameter("Operator");
if(op.equals("Addition")){
out.println("Addition is :"+(Integer.parseInt(num1)+Integer.parseInt(num2)));
}
else if (op.equals("Subtraction")){
out.println("Subtraction is : "+(Integer.parseInt(num1)-Integer.parseInt(num2)));
}
else if (op.equals("Multiplication")){
out.println("Multiplication is :"+(Integer.parseInt(num1)*Integer.parseInt(num2)));
}
else{
out.println("Division is :"+(Integer.parseInt(num1)/Integer.parseInt(num2)));
}
out.println("Calculator");
}
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {processRequest(request, response);}
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {processRequest(request, response);}
@Override
public String getServletInfo() {
return "Short description";
}
}

```

Output:-

Enter Number 1 :	3	
Enter Number 2 :	4	Addition is :7
Operator	+ reset submit	Calculator
Enter Number 1 :	6	
Enter Number 2 :	4	Subtraction is : 2
Operator	- reset submit	Calculator
Enter Number 1 :	6	
Enter Number 2 :	4	Multiplication is :24
Operator	* reset submit	Calculator
Enter Number 1 :	8	
Enter Number 2 :	4	Division is :2
Operator	/ reset submit	Calculator

AIM(B): Create a servlet for a login page. If the username and password are

correct then it says message "Hello <username>" else a message "login failed".

Code html:-

```
<html>
  <head>
    <title>Login</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
  </head>
  <form action="login" method="get" name="frm">
    Enter Username:
    <input name="uname" type="text"/>
    Enter password:
    <input name="pass" type="password"/>
    <input name="submit" type="Submit"/>
  </form>
</body>
</html>
```

Code Servlet:-

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet(name = "login", urlPatterns = {"/login"})
public class login extends HttpServlet {
    public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            String uname = request.getParameter("uname");
            String pass = request.getParameter("pass");
            if(uname.equals("user")&& pass.equals("pwd")){
                out.println("Hello <b>" + uname + "</b> welcome to java Servlet");
            }else{
                out.println("Login Failed");
            }
        }
    }
}
```

Output:-

Enter Username:

Enter password:

Hello user welcome to java Servlet

AIM(C): Create a registration servlet in Java using JDBC. Accept the details such as Username, Password, Email, and Country from the user using HTML Form and store the registration details in the database.

Code:-

```
<!DOCTYPE html>
<html>
  <head>
    <title>Practical 1 C</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
  </head>
  <body>
    <form action="login" method="post">
      <table cellpadding="1">
        <thead>
          <tr>
            <th><b>REGISTRATION FORM</b></th>
          </tr>
        </thead>
        <tbody>
          <tr>
            <td><b>First Name : </b></td>
            <td><input type="text" name="fname" value=""></td>
          </tr>
          <tr>
            <td><b>Last Name : </b></td>
            <td><input type="text" name="sname" value=""></td>
          </tr>
          <tr>
            <td><b>Zip Code : </b></td>
            <td><input type="text" name="zip" value=""></td>
          </tr>
          <tr>
            <td><b>User Name : </b></td>
            <td><input type="text" name="uid" value=""></td>
          </tr>
          <tr>
            <td><b>Password : </b></td>
            <td><input type="password" name="pwd" value=""></td>
          </tr>
          <tr>
            <td><b>Confirm Password : </b></td>
            <td><input type="password" name="pwd1" value=""></td>
          </tr>
          <tr>
            <td><b>Town : </b></td>
            <td><input type="text" name="town" value=""></td>
          </tr>
          <tr>
            <td><b>Country : </b></td>
            <td><input type="text" name="country" value=""></td>
          </tr>
        </tbody>
      </table>
    </form>
  </body>
</html>
```

```

        </tr>
        <tr>
            <td><input type="submit" value="Submit"></td>
        </tr>
    </tbody>
</table>
</form>
</body>
</html>

```

Code ServReg:-

```

import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.SQLException;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet(urlPatterns = {"/login"})
public class login extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");

        PrintWriter out = response.getWriter();
        Connection connection = null;
        PreparedStatement pst = null;

        try {

            // MySQL Connector 5.1.17 driver
            Class.forName("com.mysql.jdbc.Driver");

            connection = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/db",
                "root",
                "root"
            );

            String uid = request.getParameter("uid");
            String fname = request.getParameter("fname");
            String sname = request.getParameter("sname");

```

```
String pwd = request.getParameter("pwd");
String pwd1 = request.getParameter("pwd1");
String town = request.getParameter("town");
String country = request.getParameter("country");
String zip = request.getParameter("zip");
```

```
String sql = "INSERT INTO user_register VALUES(?,?,?,?,?,?,?,?)";
```

```
pst = connection.prepareStatement(sql);
```

```
pst.setString(1, uid);
pst.setString(2, fname);
pst.setString(3, sname);
pst.setString(4, pwd);
pst.setString(5, pwd1);
pst.setString(6, town);
pst.setString(7, country);
pst.setString(8, zip);
```

```
int rows = pst.executeUpdate();
```

```
if (rows > 0) {
    out.println("<h2>Registration Successful</h2>");
    out.println("Welcome : " + fname);
} else {
    out.println("Registration Failed");
}
```

```
} catch (ClassNotFoundException e) {
```

```
    out.println("Driver missing : " + e.getMessage());
```

```
} catch (SQLException e) {
```

```
    out.println("Database Error : " + e.getMessage());
```

```
} finally {
```

```
    try {
        if (pst != null) {
            pst.close();
        }
    }
```

```
    if (connection != null) {
        connection.close();
    }
}
```

```

    } catch (SQLException e) {
        out.println(e.getMessage());
    }
}
}
}

```

Output:-

```

mysql> USE db;
Database changed
mysql> CREATE TABLE user_register(
    -> uid CHAR(40),
    -> fname CHAR(40),
    -> sname CHAR(40),
    -> pwd CHAR(40),
    -> pwd1 CHAR(40),
    -> town CHAR(40),
    -> country CHAR(40),
    -> zipcode CHAR(40)
    -> );

```

```

mysql> SHOW VARIABLES LIKE 'PORT';
+-----+-----+
| Variable_name | Value |
+-----+-----+
| port          | 3306  |
+-----+-----+
1 row in set (0.02 sec)

```

```

mysql> select * from user_register;
+-----+-----+-----+-----+-----+-----+-----+-----+
| uid   | fname | sname | pwd  | pwd1 | town  | country | zipcode |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 2v5   | q2g54vr | v2    | 1    | 1    | e5j   | w365e   | evq25   |
| NEH   | NEHA  | GIRI  | 1    | 1    | MUMBAI | INDIA   | 12346   |
+-----+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.02 sec)

```

REGISTRATION FORM

First Name :	NEHA
Last Name :	GIRI
Zip Code :	12346
User Name :	NEH
Password :	•
Confirm Password :	• 
Town :	MUMBAI
Country :	INDIA
Submit	

Registration Successful

Welcome : NEHA

